ElderGuard: What Changed

A simple before-and-after summary of the backend work | November 7, 2026

What is ElderGuard? Cameras watch rooms in a home for the aged. An AI model looks for falls, and when it sees one, staff get an alert on their dashboard. The **backend** is the behind-the-scenes part that runs the cameras, saves alerts and controls who can see what.

The short version: the system already worked, but a real fall could sometimes go unreported, passwords were written inside the code, and unusual input could cause errors. Those problems are now fixed and tested.

3	6	53	248 / 249
ways a real fall could go unreported, now fixed	security problems fixed	bugs found by testing, all fixed	test checks now pass

1. Making sure real falls are not missed

This is the most important part, because a missed alert could mean someone stays on the floor without help.

Topic	Before	Now
Two falls close together	After any alert, that camera ignored new falls for 60 seconds. If a real fall came just after a false alarm, nobody was told.	Every fall the system sees happening raises an alert. Only a repeat of the same person, already down in the same spot, is ignored.
Database problem	If an alert could not be saved, it was lost and only a console message appeared.	The alert is kept and retried every 5 seconds, with the original time of the fall.
Camera program crash	The camera stopped working but still showed as online, with a frozen picture.	It shows as offline with the reason, then restarts by itself after 5 seconds.

2. Keeping accounts and private information safe

Think of these as changing the locks and no longer leaving the spare key under the doormat.

Topic	Before	Now
Passwords in the code	The database password and the secret key for logins were written inside a code file.	They are in a separate private file (**backend/.env**) that is kept out of shared code.
Default admin password	A new system always started with the same easy-to-guess admin password.	A new system creates a random password, shown only once.
Temporary passwords	The rule to change a temporary password could be skipped.	Until it is changed, that account can do almost nothing.
Bathroom privacy	A bathroom camera could be set with no blur at all.	Bathroom cameras always blur the whole picture. This cannot be turned off.

Topic	Before	Now
Camera links	Regular staff could see camera links, which can contain camera passwords.	Only admins can see them.
Server logs	Login tokens and camera links could be written into the server's log.	They now appear as *** in the log.

3. Serious problems found by testing

After those fixes, I tested the backend the way a QA tester would, trying to break it on purpose.

Topic	Before	Now
Removed staff	A deactivated staff member's open screen kept receiving alerts and live video.	They are cut off: video within seconds, and alerts from the very next one.
Incident records	An old alert showed who is in the room **now**, so records changed when a resident moved.	New alerts save who was in the room **at that moment**.
Last admin	The only admin could remove their own admin rights, leaving nobody in charge.	This is refused until there is another admin.
Server errors	Unusual input, like an empty required field or a huge number, caused a server error in 24 of the tests.	The server gives a clear message instead. The final test run had no server errors.

4. Everyday behavior

These affect how the system feels to use each day.

Topic	Before	Now
Same-time clicks	Several staff could acknowledge the same alert at once, so the record of who responded could be wrong.	Only the first one counts, and the others are told it was already handled.
Falls today count	False alarms and test alarms were counted as real falls.	Only real falls count. Test alarms are now clearly marked.
Resident search	Searching "maria" did not find "Maria Santos".	Upper and lower case no longer matter.
Wrong login found	Someone's email matching another person's username could block that person from logging in.	A username match always comes first, and two accounts cannot share an email.
Guessing passwords	Nothing slowed down repeated guessing across different accounts.	After 30 failed logins in a minute from one address, it must wait.
Logging out	There was no real logout, and old logins kept working after a password change.	Logout ends the login on the server, and a password change ends older logins.
Wrong information	Blank names, invalid emails and an age of 500 were all accepted.	These are refused with a clear message.

5. Test results

I added more checks as I fixed things, which is why the total went up. The tests used separate temporary databases, so your real records were not changed.

Test score	First test	Final test
Checks passed	126 of 179	**248 of 249**
Server errors	30	**0**
Checks that failed	53	**1** (explained below)

The 1 remaining failure: a check I added for the new login limit. A trick gets around the limit, but only for someone already using the server's own computer. It does not work from other computers.

6. What is not solved yet

- **The AI still makes mistakes.** It missed the fall in one test video (**b7.mp4**), and it can mistake sitting for falling. Your 24 deleted photos were this kind of false alarm. It needs more training examples.
- **More false alarms may reach staff.** The old 60-second pause also hid false alarms. Removing it protects real falls, so better training matters even more now.
- **Nobody is chased up.** Alerts only appear on open dashboards, and nothing happens if nobody responds.
- **Old alerts have no room record.** Alerts made before this update still show the room's current residents.
- **Not tested yet:** the website (frontend), real webcams or network cameras, running for days at a time, and the database being fully shut down.

7. What you need to do

- **Restart the backend.** The changes start working then, and it updates your database by itself. Your existing alerts are kept. Everyone will need to log in again.
- **Change your PostgreSQL password,** then update it in **backend/.env**. The old one was stored in the code.
- **Delete the 24 false alerts,** if you still want them gone. Your database still had them at the last check.
- **Check the website.** It may need small updates to use the new logout and new messages.
- **Want the /docs page while developing?** It is now off. Add **ELDERGUARD_DOCS=1** to **backend/.env**.